

Module-2

Session-2

Q1. Install GCC on your computer and verify the installation by running `gcc --version` in your terminal or command prompt.

Ans. Installing and Verifying GCC

To compile C programs, you need the **GCC (GNU Compiler Collection)** compiler.

- **Windows:** The easiest way is to download **MinGW-w64** (often bundled with DevC++) and add its bin folder to your System Environment Variables (Path).
- **Mac:** Open your terminal and run `xcode-select --install`.
- **Linux:** Run `sudo apt install build-essential` in your terminal.

Verification

Once installed, open your Command Prompt (Windows) or Terminal (Mac/Linux) and type:

```
gcc --version
```

If successful, you will see output details showing the installed version of GCC (e.g., gcc (MinGW-w64...) 11.2.0).

Q2. Open VS Code or DevC++ and create a new file named `hello.c`. Write a C program that prints 'Hello, Instagram World!' to the console.

Ans. This is the raw, functional C program without any explanations. If you just want to run the code, this is all the computer needs.

```
#include <stdio.h>

int main() {

    printf("Hello, Instagram World!\n");

    return 0;

}
```

Q3. Identify and label the header, main function, and return statement in your `hello.c` program using comments.

Ans. This version uses **comments** (`//`). Comments are completely ignored by the GCC compiler; they are written strictly for humans to understand the anatomy of the program.

```
// 1. THE HEADER FILE
```

```
// #include <stdio.h> tells the computer to load the Standard Input-Output library.
```

```
// Without this, the computer won't understand what "printf" means.
```

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```
#include <stdio.h>

// 2. THE MAIN FUNCTION

// int main() is the mandatory entry point of every C program.

// The computer looks for this exact name to know where to start reading your instructions.

int main() {

    // This is the action statement that prints your message to the screen.

    printf("Hello, Instagram World!\n");

    // 3. THE RETURN STATEMENT

    // return 0; sends a signal back to your operating system saying:

    // "Everything went perfectly, the program finished without crashing."

    return 0;

}
```

Q4. Compile your hello.c file using the GCC command line (gcc hello.c -o hello) and run the output file to display the message in your terminal.
Hint: On Windows, the output file will be hello.exe; on Linux/Mac, it will be hello.

Ans.

Now, let's translate this code into a runnable application using your terminal.

Step A: Navigate to your file

cd Documents/C_Programs

Step B: Compile the code

Run the GCC compilation command. This takes your human-readable code and builds an executable binary file:

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```
gcc hello.c -o hello
```

Step C: Run the executable

Depending on your operating system, type the following command to execute your new program:

- **On Windows:**

```
hello.exe
```

- **On Mac / Linux:**

```
./hello
```

Expected Output:

```
Hello, Instagram World!
```